

Based on the DSBDA (Data Science and Big Data Analytics) lab index provided, here are common viva questions and answers categorized by experiment topic:

1. Data Wrangling & Descriptive Statistics

Q: What is Data Wrangling?

- **A:** It is the process of cleaning, transforming, and mapping raw data into a usable format for analysis. It includes handling missing values, filtering, and merging datasets.

Q: Explain the difference between Mean, Median, and Mode.

- **A:**
 - **Mean:** The average of all data points.
 - **Median:** The middle value when data is sorted.
 - **Mode:** The value that appears most frequently.

Q: What is Standard Deviation?

- **A:** It measures the amount of variation or dispersion in a set of values. A low SD means data points are close to the mean; a high SD means they are spread out.

2. Regression Analysis (Linear & Logistic)

Q: When do we use Linear vs. Logistic Regression?

- **A:**
 - **Linear Regression:** Used for predicting continuous numerical values (e.g., predicting house prices in the Boston dataset).
 - **Logistic Regression:** Used for classification problems to predict discrete outcomes (e.g., Yes/No, or whether a user clicked an ad).

Q: What is a Confusion Matrix?

- **A:** A table used to evaluate the performance of a classification model. It displays True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN).

3. Machine Learning (Naive Bayes)

Q: What is the fundamental principle of Naive Bayes?

- **A:** It is based on **Bayes' Theorem** and assumes that the presence of a particular feature in a class is unrelated to the presence of any other feature (Independence Assumption).

Q: Why is it called "Naive"?

- **A:** Because it assumes all features in the dataset are independent of each other, which is rarely true in real-world data.

4. Text Analytics

Q: What is Tokenization and Lemmatization?

- **A:**
 - **Tokenization:** Breaking down a sentence into individual words or "tokens."
 - **Lemmatization:** Reducing a word to its meaningful base form (e.g., "running" becomes "run"). Unlike stemming, it considers the context.

Q: Explain TF-IDF.

- **A:** It stands for **Term Frequency-Inverse Document Frequency**. It evaluates how important a word is to a document in a collection. It increases proportionally to the number of times a word appears in the document but is offset by the frequency of the word in the entire corpus.

5. Data Visualization

Q: What is a Box Plot used for?

- **A:** It is used to show the distribution of data based on a five-number summary: minimum, first quartile (Q1), median, third quartile (Q3), and maximum. It is excellent for identifying **outliers**.

Q: When would you use a Histogram?

- **A:** To visualize the frequency distribution of a single continuous numerical variable (e.g., the distribution of 'fare' in the Titanic dataset).

Q: Why use Seaborn over Matplotlib?

- **A:** Seaborn is built on top of Matplotlib and provides a high-level interface for drawing attractive statistical graphics with fewer lines of code and better default styles.

1. Data Visualization (Titanic & Iris Datasets)

Q: What is the purpose of a Box Plot?

- **A:** A box plot (or whisker plot) is used to show the distribution of quantitative data. It highlights the median, quartiles, and potential outliers. In your assignment, it helps compare the age distribution between genders.

Q: What does a Histogram represent?

- **A:** A histogram represents the frequency distribution of a continuous variable (like ticket 'fare'). It shows how many data points fall within specific ranges (bins).

Q: How do you identify an outlier in a dataset?

- **A:** In a box plot, outliers are usually represented as individual points beyond the "whiskers." Mathematically, they are values that fall below $(Q1 - 1.5 \times IQR)$ or above $(Q3 + 1.5 \times IQR)$ (where IQR is the Interquartile Range).

2. Map-Reduce (Word Count Assignment)

Q: Explain the basic flow of a Map-Reduce program.

- **A:** It consists of two main phases:
 - **Map:** Processes input data and produces key-value pairs (e.g., <word, 1>).
 - **Reduce:** Aggregates the values associated with the same key (e.g., summing all 1s for a specific word to get the total count).

Q: Why use Map-Reduce for large datasets?

- **A:** It allows for distributed processing across multiple nodes in a cluster, making it much faster to process terabytes of data than a single machine could.

3. Apache Spark & SCALA

Q: What is Apache Spark, and how does it differ from Hadoop Map-Reduce?

- **A:** Spark is a distributed computing framework. Unlike Hadoop Map-Reduce, which writes data to the disk after every step, Spark performs **in-memory processing**, making it significantly faster for iterative tasks.

Q: What is an RDD in Spark?

- **A:** RDD stands for **Resilient Distributed Dataset**. It is the fundamental data structure of Spark—an immutable, fault-tolerant collection of objects that can be operated on in parallel.

4. Machine Learning (Movie Recommendation)

Q: What is the difference between Content-Based and Collaborative Filtering?

- **A:**
 - **Content-Based:** Recommends items similar to those a user liked before (based on item features like genre/actors).
 - **Collaborative Filtering:** Recommends items based on the preferences of similar users ("Users who liked X also liked Y").

5. Data Analysis (COVID-19 Mini Project)

Q: What are the primary steps in Exploratory Data Analysis (EDA)?

- **A:**
 - **Data Cleaning:** Handling missing values or incorrect entries.
 - **Univariate Analysis:** Looking at one variable at a time (e.g., total vaccinations).
 - **Bivariate/Multivariate Analysis:** Looking at relationships between variables (e.g., vaccination rate vs. state population).

Q: How do you handle missing values in a dataset like the vaccine CSV?

- **A:** Common methods include **dropping** rows with missing values, **imputing** them (replacing with mean, median, or mode), or using algorithms that handle missing data natively.

Part 1: Python Libraries (Pandas, NumPy, Matplotlib, Seaborn, Sklearn)

1) What is Pandas library in Python?

Pandas is an open-source Python library used for data manipulation, analysis, and cleaning.

2) List some key features of Pandas.

- **DataStructures:** Two primary data structures: DataFrame (2D, labeled) and Series (1D).
- **Data Cleaning:** Functions for handling missing data and removing duplicates.
- **Manipulation:** Easy merging, joining, and reshaping of datasets.
- **Grouping:** Powerful functionality for split-apply-combine operations.
- **I/O Tools:** Loading data from CSV, SQL, JSON, and Excel.
- **Time Series:** Extensive support for time-stamped data.

3) What is Numpy Library in Python?

NumPy (Numerical Python) is an open-source Python library used for scientific computing and numerical operations.

4) What is matplotlib library?

Matplotlib is a low-level Python library used for creating static, animated, and interactive visualizations.

5) What is difference between seaborn library and matplotlib lib?

Matplotlib is a low-level plotting library for custom graphs, while Seaborn is a high-level library that provides easier and more attractive statistical visualizations.

6) Is Sklearn and Scikit Learn are same library? What is use of it in data science?

Yes, **Scikit-learn** and **sklearn** refer to the same library—**sklearn** is the package name used in import statements. It is widely used in data science for **machine learning tasks** such as classification, regression, clustering, and data preprocessing.

7) What are functions comes in Pandas and Numpy library?

Pandas Functions

1. Data Inspection

- `head()` – View first 5 rows
- `tail()` – View last 5 rows
- `info()` – Summary of DataFrame
- `describe()` – Statistical summary

2. Data Selection & Indexing

- `loc[]` – Label-based selection
- `iloc[]` – Position-based selection

3. Data Cleaning

- `isnull()` / `notnull()` – Check missing values
- `dropna()` – Remove missing values
- `fillna()` – Fill missing values
- `drop_duplicates()` – Remove duplicates

4. Data Manipulation

- `drop()` – Remove rows/columns
- `rename()` – Rename labels
- `sort_values()` – Sort data
- `astype()` – Change data type

5. Data Aggregation

- `groupby()` – Group data
- `agg()` – Aggregate functions
- `value_counts()` – Frequency count

6. Data Combining

- `merge()` – Join datasets (SQL-style)
- `concat()` – Concatenate datasets

NumPy Functions

1. Array Creation

- `array()` – Create array
- `zeros()` – Array of zeros
- `ones()` – Array of ones
- `arange()` – Range of values
- `linspace()` – Evenly spaced values

2. Mathematical Operations

- `mean()` – Average
- `median()` – Median
- `std()` – Standard deviation
- `sum()` – Sum
- `min()` / `max()` – Minimum / Maximum

3. Array Manipulation

- `reshape()` – Change shape
- `resize()` – Resize array
- `flatten()` – Convert to 1D
- `transpose()` – Transpose matrix

4. Indexing & Filtering

- Boolean indexing
- Slicing (`arr[1:5]`)

5. Linear Algebra

- `dot()` – Matrix multiplication
- `linalg.inv()` – Inverse of matrix
- `linalg.eig()` – Eigenvalues & vectors

8) What is data frame in Python?

A DataFrame is a two-dimensional, size-mutable, heterogeneous tabular data structure with labeled axes (rows and columns). It is conceptually similar to a SQL table or an Excel spreadsheet. [7, 8]

9) How to find duplicates in python? (Python Command)

// Find Duplicate Rows

df.duplicated()

- **Returns a Boolean Series**
- **True** → duplicate row
- **False** → unique row

// Remove Duplicate Rows

df.drop_duplicates()

Removes duplicate rows from the DataFrame

Returns a new DataFrame with only unique values

10) What is use of describe command?

calculates summary statistics of the DataFrame, including count, mean, standard deviation, minimum, maximum, and quartile values (25%, 50%, 75%) for numerical columns.

Part 2: Machine Learning & Visualization

11) Which are Naive Bayes classification algorithm used in python? provides several Naive Bayes implementations:

- : For numerical features following a normal distribution.
- : For discrete counts (text classification).
- : For binary/boolean features. [17, 19]

12) What is significance of Confusion matrix?

A confusion matrix is used to evaluate the performance of a classification model. It shows the true positives, true negatives, false positives, and false negatives, helping to understand how often the model confuses classes. [15]

13) What is TP, TN, FP, FN in confusion matrix?

- **TP (True Positive):** Predicted Positive, Actually Positive.
- **TN (True Negative):** Predicted Negative, Actually Negative.
- **FP (False Positive):** Predicted Positive, Actually Negative (Type I error).
- **FN (False Negative):** Predicted Negative, Actually Positive (Type II error).

14) What is recall? Recall (Sensitivity) measures the proportion of actual positives that were correctly identified:

$$\text{Recall} = \frac{TP}{TP + FN}$$

15) What is precision? Precision measures the proportion of predicted positives that were actually positive:

$$\text{Precision} = \frac{TP}{TP + FP}$$

16) What is f1 score? The F1 score is the harmonic mean of precision and recall, providing a balanced measure of both, especially when classes are imbalanced:

$$\text{F1 Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

17) What is need of data visualization in data science? Data visualization transforms huge datasets into graphs (histograms, scatter plots, etc.), making complex data understandable, accessible, and aiding in identifying patterns, trends, and outliers. [22]

18) What is outlier? An outlier is a data point that differs significantly from other observations in a dataset. [8]

19) When to use histogram and pie chart?

- **Histogram:** Used to visualize the distribution of a continuous numerical variable.
- **Pie Chart:** Used to represent the composition of a whole, showing proportions of categorical data (best with fewer categories). [23, 24]

20) What are the challenges in Big data visualization?

- **Volume:** Too many data points causing clutter and slow rendering.
- **Variety:** Handling different data types (text, images, graphs).
- **Velocity:** Real-time visualization requirements. [8, 15, 25]

21) What is joint plot, dist plot?

- **Distplot (Distribution Plot):** Shows the univariate distribution of a single variable, combining a histogram with a Kernel Density Estimate (KDE) plot.

- **Jointplot:** Used to visualize the relationship (scatter plot) and marginal distributions (histograms) between two variables simultaneously. [[12](#), [23](#)]

22) What are tools used for data visualization?

Matplotlib, Seaborn, Plotly, Tableau, Power BI, and Bokeh.

Part 3: Data Preprocessing, Big Data & Scala

23) What is data Wrangling? Data Wrangling is the process of cleaning, structuring, and transforming raw data into a usable format for analysis, such as handling missing values, changing data types, and restructuring data. [[8](#)]

24) What is data transformation? Data transformation involves changing the format, structure, or values of data, such as normalization, standardization, encoding categorical variables, or logarithmic transformations. [[7](#)]

25) What is use of standard scaler function in python? from standardizes features by removing the mean and scaling to unit variance (Z-score normalization), crucial for algorithms sensitive to data scale (like SVM, KNN). [[19](#)]

26) What is hadoop? Apache Hadoop is an open-source framework designed to store and process large-scale data (Big Data) in a distributed environment using clusters of computers. [[25](#)]

27) What is HDFS and Map reduce?

- **HDFS (Hadoop Distributed File System):** The storage component of Hadoop, which stores data across multiple nodes in a cluster.
- **MapReduce:** The processing component of Hadoop, which breaks down large tasks into smaller, parallel tasks (Map) and then combines results (Reduce). [[6](#), [25](#)]

28) What are the component of Hadoop Ecosystem?

- **HDFS:** Storage
- **MapReduce:** Processing
- **YARN:** Resource Management
- **Hive:** SQL-like querying
- **Pig:** High-level scripting
- **Spark:** Fast in-memory processing. [[8](#)]

29) What is Scala? Scala (Scalable Language) is a high-level, strongly-typed programming language that combines functional programming and object-oriented programming, designed to run on the Java Virtual Machine (JVM). [6]

30) What are features of Scala?

- Functional and Object-Oriented blend.
- Type Inference.
- Lazy Evaluation.
- High Scalability (runs on JVM).
- Built-in support for concurrency. [6]

31) How scala is different from java?

- Scala supports functional programming (first-class functions, immutability), whereas Java is primarily object-oriented (though Java 8+ introduced functional concepts).
- Scala has more concise syntax compared to Java.
- Scala supports type inference, while Java requires explicit typing. [6]

32) List application of Scala.

- Big data processing (Apache Spark is built in Scala).
- Distributed computing systems.
- High-performance web services.
- Functional programming applications. [6]

33) What is data science? Data science is a multidisciplinary field that uses scientific methods, processes, algorithms, and systems to extract knowledge and insights from structured and unstructured data. [26]

34) What is Big data? Big Data refers to datasets that are too large, fast, or complex for traditional data processing software to handle effectively. [8]

35) What are the characteristics of Big data? Commonly known as 5 Vs: Volume, Velocity, Variety, Veracity, and Value. [6]

36) List phases in data science life cycle?

1. Business Understanding
2. Data Acquisition
3. Data Preparation/Wrangling
4. Data Exploration (EDA)
5. Modeling
6. Evaluation
7. Deployment. [2]

37) What is Central tendency? Central tendency is a statistical measure that represents the center or typical value of a data set, including Mean, Median, and Mode. [7]

38) What is dispersion? Dispersion measures how spread out the data points are from the central tendency (e.g., Variance, Standard Deviation, Range). [6]

39) What is mean, mode, mid range, median? Calculate same for 10,22,13,10,21,43,77,21,10

- **Data:** 10, 10, 10, 13, 21, 21, 22, 43, 77 (Sorted)
- **Mean:** $\frac{10+22+13+10+21+43+77+21+10}{9} = \frac{227}{9} \approx 25.22$
- **Median:** Middle value (5th) = 21
- **Mode:** Most frequent value = 10 (appears 3 times)
- **Mid Range:** $\frac{\text{Max} + \text{Min}}{2} = \frac{77+10}{2} = 43.5$

40) What is Variance? Variance is the average of the squared differences from the mean, measuring the spread of the data. [27]

41) What is Standard Deviation? Standard Deviation is the square root of variance, providing a measure of dispersion in the same units as the data. [27]

42) What is mean by posterior probability on naive bayes theorem? Posterior probability is the conditional probability of a class C (label) given the observed data X , calculated as $P(C|X)$. It is the final answer Naive Bayes predicts, updated from the prior probability. [19]

43) What is mean by likelihood probability on naive bayes theorem? Likelihood is the probability of observing data X given that the class C is true, represented as $P(X|C)$. It measures how well the data fits the class. [19]

44) How we can deal with Missing values? What way to deal with missing value or null value.

- **Drop:** to remove rows or columns with null values.
- **Impute:** to replace NaN with a specific value, mean, median, or mode.
- **Interpolate:** Filling values using techniques like linear interpolation. [7]

45) What is NLTK?NLTK (Natural Language Toolkit) is a leading Python library for Natural Language Processing (NLP), providing tools for tokenization, stemming, lemmatization, and parsing. [1]

46) What is Tokenization in NLP?Tokenization is the process of breaking down a text into smaller units, such as words or sentences, called tokens. [8]

47) What is Stemming?Stemming is the process of reducing a word to its root form (e.g., "running" becomes "run"), often by cutting off suffixes. [1]

48) What is Lemmatization?Lemmatization reduces words to their base or dictionary form (lemma) using a vocabulary and morphological analysis (e.g., "better" becomes "good"). [1]

49) What is Corpus in NLP?A corpus is a large, structured set of texts used for analyzing and training language models. [8]

50) What is Spark framework?Apache Spark is an open-source, distributed computing system that provides high-speed, in-memory data processing, offering APIs in Java, Scala, Python, and R.

Steps to run Scala program in windows with Spark framework:

1. **Copy file:** Copy your file to the directory (e.g.,).
2. **Open Command Prompt:** Open Command Prompt (CMD) and navigate to the folder, or type in the address bar of the Spark folder.
3. **Start Spark-shell:** Type and press enter.
4. **Run file:** To run your file, type: (e.g.,).